

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY  
SCHOOL OF INFORMATION AND COMMUNICATION  
TECHNOLOGY



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## FINAL PROJECT REPORT

Topic: **Parallelization Methods for CNN-Based Computer Vision Inference**

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# Abstract

This report presents the final structure for the project *Parallelization Methods for CNN-Based Computer Vision Inference*. The final scope focuses on one implemented method: a YOLO/OpenMPI video people-counting pipeline that parallelizes inference at task level over video frames and image tiles.

The report uses measured values already recorded in the project reports. The YOLO pipeline runs on a three-MacBook physical cluster with C++17 and OpenMPI. A video is decomposed temporally into frames and spatially into tiles; each frame-tile pair becomes an MPI task. The final reported pipeline emphasizes static block-cyclic scheduling, rank-local YOLO workers, coordinate remapping, duplicate removal, global NMS, CSV output, and per-rank timing.

The recorded YOLO experiments verify serial-versus-MPI correctness on 30 frames with zero mismatched frames. The detector accuracy experiment on 300 MOT17-derived frames reports mean absolute count error 7.983 and predicted average count 8.223 versus ground-truth average 16.207. The input-size experiment selects  $N = 600$  frames because the 12-process runtime is 123.667 seconds, satisfying the required two-to-three-minute workload. On  $2N = 1200$  frames, the 12-process weighted 4/6/2 run reaches 129.852 seconds and 2.662x wall-clock speedup. Weighted static placement improves the idle-gap indicator from 0.466 to 0.235.

# Chapter 1

## Introduction

Computer vision inference is a useful setting for studying parallel programming because the workload is computationally heavy, data-rich, and naturally decomposable. A CNN-based detector can be applied repeatedly across frames and image regions. These repeated operations expose the main concerns of a parallel system: decomposition, mapping, communication, load balancing, correctness, and speedup.

This project focuses on CNN-based inference rather than model training. The main implemented application is video people counting with YOLO. Given an input video, the system predicts people bounding boxes and frame-level counts. The expensive work is detector inference over many frames and image tiles. The parallel contribution is the C++17/OpenMPI runtime around the detector: it creates tasks, schedules them across MPI ranks, launches local detector workers, gathers detections, removes duplicates, and writes experiment artifacts.

The report separates two types of correctness. Model accuracy means whether YOLO counts match human ground truth from MOT17. Parallel correctness means whether the MPI implementation produces the same result as the serial version of the same pipeline. A detector can undercount in crowded scenes while the parallel implementation remains correct. The measured results from the repository show this distinction clearly: serial and MPI counts match in the correctness test, while detector-versus-ground-truth accuracy remains limited on crowded MOT17 frames.

The main objectives are:

- define CNN-based computer vision inference as a parallel workload;
- present task parallelism for YOLO video inference;
- report measured YOLO correctness, input-size, scheduling, granularity, speedup, and weighted-placement results from the repository;
- summarize the accuracy and performance trade-offs of the final YOLO MPI pipeline.

## Chapter 2

# Problem Description

### 2.1 CNN-Based Computer Vision Inference

The problem is to execute CNN-based video inference faster by distributing repeated detector work across MPI processes. The program applies a fixed pretrained model; it does not train or update model weights. The parallel system must preserve the output of the serial inference pipeline while reducing runtime or revealing the cost of communication.

### 2.2 YOLO Video People Counting

The input video is a finite sequence of frames:

$$V = \{F_1, F_2, \dots, F_N\}. \quad (2.1)$$

The target output for each frame is the number of detected people after post-processing:

$$\text{count}(F_i) = |\{b \mid b \text{ is a final person bounding box in } F_i\}|. \quad (2.2)$$

The detector backend is a pretrained YOLO11n model through a local Python worker process [4]. YOLO11 follows the usual one-stage detector structure with a feature-extraction backbone, a feature-fusion neck, and multi-scale detection heads, as shown in Figure 2.1. The MPI program does not split YOLO neural-network kernels. Instead, MPI distributes repeated YOLO calls over frame-tile tasks.

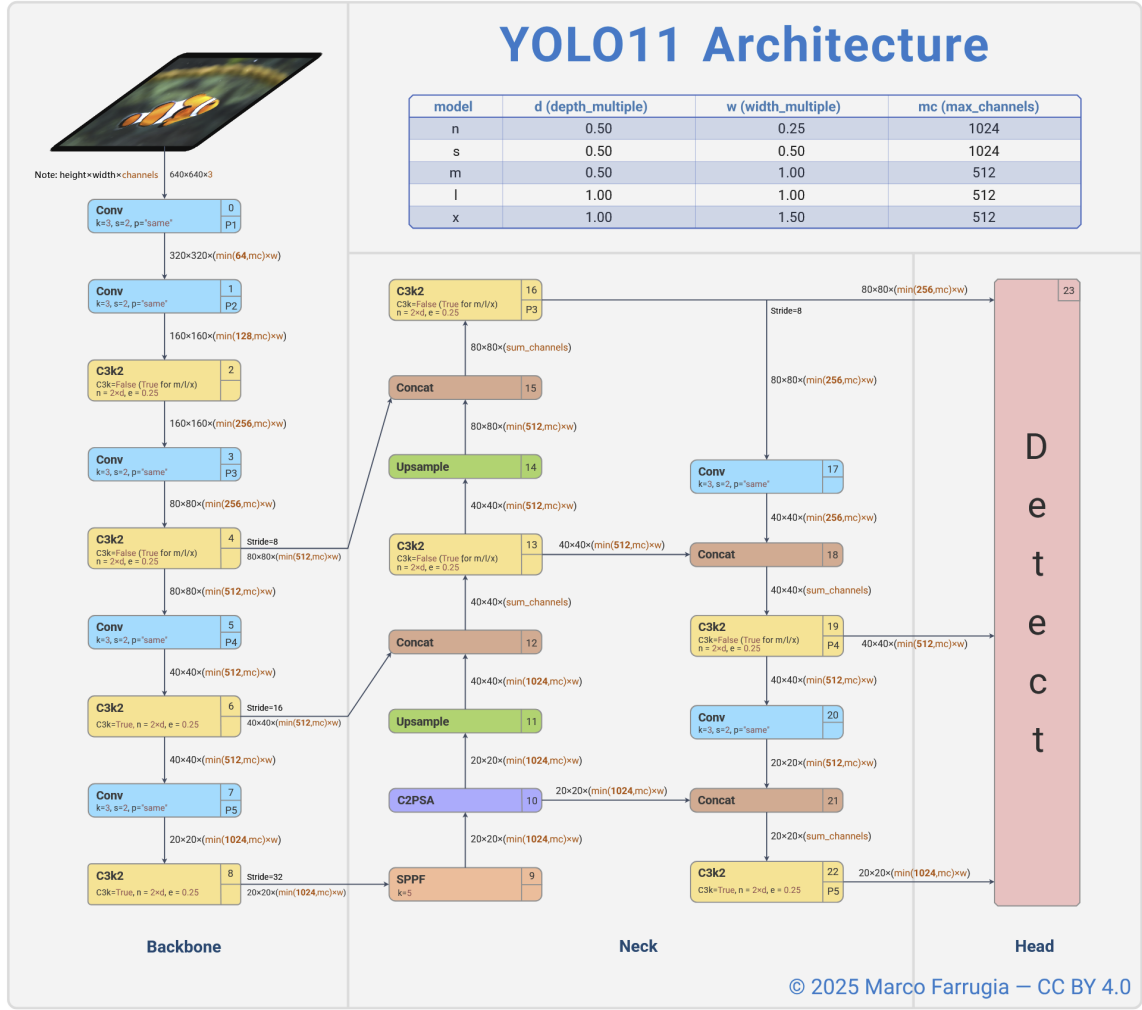


Figure 2.1: YOLO11 object-detection architecture used as the detector backend in this project. Source: Farrugia, CC BY 4.0, Zenodo DOI 10.5281/zenodo.17508018 [1].

For a tile grid with  $R$  rows and  $C$  columns, frame  $F_i$  is decomposed into:

$$F_i \rightarrow \{T_{i,1}, T_{i,2}, \dots, T_{i,R \times C}\}. \quad (2.3)$$

A task is one frame-tile inference:

$$task(i, j) = T_{i,j}, \quad |\mathcal{T}| = N \times R \times C. \quad (2.4)$$

Rank 0 converts tile-local detections back to full-frame coordinates, applies tile ownership filtering, removes duplicates, and writes final frame counts.

## 2.3 Parallel-Programming Focus

The final report studies one implemented MPI pipeline in depth instead of splitting attention across two methods. The YOLO pipeline exercises the following parallel-programming concepts:

Table 2.1: Parallel-programming concepts in the YOLO pipeline

| Concept            | YOLO MPI implementation                                                            |
|--------------------|------------------------------------------------------------------------------------|
| Parallelism level  | Task parallelism across repeated YOLO inference calls                              |
| Decomposition      | Hybrid temporal-spatial decomposition into frame-tile tasks                        |
| Mapping            | 1D flattened task list with static block-cyclic ownership                          |
| Communication      | Rank-local inference followed by centralized detection and metric gathering        |
| Load balancing     | Tile granularity and weighted process placement                                    |
| Correctness target | MPI frame counts match serial frame counts after the same post-processing pipeline |

## 2.4 Correctness Targets

The serial YOLO baseline processes the same frame-tile task list without MPI distribution and applies the same post-processing pipeline. Parallel correctness is:

$$\Delta_i = |\text{count}_{\text{parallel}}(F_i) - \text{count}_{\text{serial}}(F_i)|. \quad (2.5)$$

The correctness experiment reports mismatched frames, maximum count error, and mean count error. Detector accuracy against MOT17 ground truth is evaluated separately because model error and parallelization error are different questions.

## Chapter 3

# Parallelization Methods

### 3.1 Task Parallel YOLO Inference

The YOLO pipeline uses task-level parallelism over independent frame-tile inference tasks. The implementation is centered on `src/yolo_mpi_cpp.cpp` in the repository. Each MPI rank runs the same C++ program, and rank-local detector work is delegated to a long-lived Python YOLO worker.

#### 3.1.1 Temporal-Spatial Decomposition

The input video is first decomposed by time into frames and then by space into image tiles. If the video has  $N$  frames and each frame is split into an  $R \times C$  grid, the total number of tasks is  $NRC$ . Figure 3.1 compares the original frame with 2x2 and 5x4 tiling. The 2x2 grid keeps detector accuracy close to the original-frame baseline because it introduces fewer artificial tile borders, so fewer people are cut by tile edges before the final merge stage. Larger grids create more tasks and can improve scheduling flexibility, but they increase duplicate detections near tile borders and increase post-processing work.

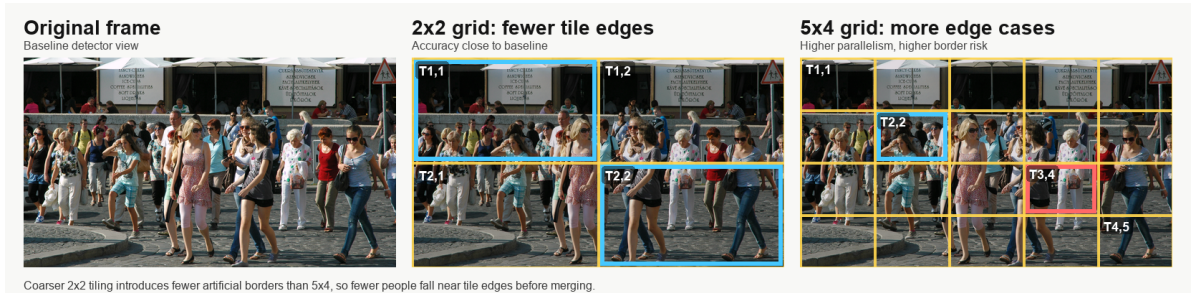


Figure 3.1: Hybrid temporal-spatial decomposition for YOLO video inference. The public-domain crowd frame is from Wikimedia Commons [2]; the overlays show why coarse 2x2 tiling preserves accuracy better than finer 5x4 tiling while still creating independent frame-tile tasks.

#### 3.1.2 1D Mapping and Static Scheduling

All frame-tile tasks are flattened into a one-dimensional list. For task index  $i$ , chunk size  $k$ , and  $P$  ranks, static block-cyclic scheduling assigns:

$$\text{owner}(\text{task}_i) = \left\lfloor \frac{i}{k} \right\rfloor \bmod P. \quad (3.1)$$



Static scheduling has low dispatch overhead because ranks can determine their own tasks locally. The measured repository results show that this advantage matters for the 600-frame 5x4 tiled YOLO workload.

### 3.1.3 Communication and Merging

After local inference, rank 0 gathers detections and rank metrics from all ranks. It then remaps tile-local boxes into frame coordinates, applies tile-owner filtering, runs global non-maximum suppression, and computes per-frame counts. This centralized merge is necessary because people near tile boundaries may be detected by multiple tiles.

### 3.1.4 YOLO Algorithm Sketch

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**Algorithm 1** Static MPI YOLO people-counting pipeline

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- 1: Build task list  $\mathcal{T}$  from  $N$  frames and an  $R \times C$  tile grid.
  - 2: **for** each MPI rank  $r$  in parallel **do**
  - 3:   Select tasks whose static owner is  $r$ .
  - 4:   Run local YOLO inference for selected tasks.
  - 5:   Store detections and timing metrics.
  - 6: **end for**
  - 7: Gather serialized detections and rank metrics to rank 0.
  - 8: Rank 0 remaps boxes, removes duplicates, computes frame counts, and writes CSV outputs.
- 

## 3.2 Weighted Process Placement

The physical cluster is heterogeneous: the three machines do not have identical CPU throughput. Equal rank placement is simple, but it can leave faster machines waiting for slower ones. Weighted process placement keeps the same static task-mapping algorithm while changing the number of MPI ranks launched on each host. Stronger machines receive more ranks and therefore more positions in the block-cyclic task stream; the slower machine receives fewer ranks.

The report evaluates uniform 4/4/4 placement and weighted placements such as 3/6/3 and 4/6/2. This isolates process placement from the scheduling algorithm: if runtime and idle gap improve, the reason is the host-to-rank assignment rather than a different task dispatch policy.

## 3.3 Metric Collection

Each run records both output correctness and timing behavior. The most important timing quantities are wall-clock runtime, runtime without communication, per-rank computation time, communication time, and the idle-gap indicator. These values are needed because speedup alone can hide the reason for poor performance. For this workload, a run can have many tasks and still be slow if rank 0 spends too much time collecting detections or if a slower node receives too much work.

## 3.4 Load Balancing and Metrics

The report uses:

$$T_r = T_{compute,r} + T_{comm,r} + T_{idle,r}, \quad (3.2)$$

and the idle-gap indicator reported by the project scripts. The course criterion treats the run as balanced when the idle gap is at most 0.25. The measured YOLO results show that equal process placement does not meet this criterion, while the best weighted static placement does.

## Chapter 4

# Experimental Setup

### 4.1 Repository Evidence

The report is based on the local repository `yolo-mpi-people-count`. The strongest source for measured YOLO values is `reports/final_report_hust_style.md`, with additional full-cluster discussion in `reports/additional_experiments_20260623.md`. The artifact branch also provides compact CSV files under `reports/artifacts/`; the YOLO MPI CSV artifacts have been copied into this report repository under `artifacts/`.

### 4.2 Physical Cluster

The project uses three physical MacBook machines connected through a local network. The benchmark mode uses CPU execution to match the course emphasis on MPI processes and processor assignment. GPU/MPS execution is kept for demonstration and live-camera use rather than formal CPU speedup measurement.

Table 4.1: Physical cluster roles

| Host   | Machine type   | Role in experiments                                                              |
|--------|----------------|----------------------------------------------------------------------------------|
| master | MacBook Air M4 | Coordinates execution, gathers results, stores outputs, and can run local tasks. |
| node1  | MacBook Pro M4 | Worker host; measured results suggest it can sustain more assigned ranks.        |
| node2  | MacBook Air M2 | Worker host; weighted placement assigns fewer ranks to reduce imbalance.         |

### 4.3 Software and Data

The implementation uses C++17, OpenMPI, Python helper scripts, and a pretrained Ultralytics YOLO detector. The main dataset source is MOT17-derived video data [3]. Compact MOT17 subsets are used for quick correctness and granularity experiments; longer sequence segments are used for the required input-size and speedup runs.

Table 4.2: Main YOLO experimental configuration

| Item               | Setting                                    |
|--------------------|--------------------------------------------|
| Detector           | Pretrained YOLO11n, person-class filtering |
| Parallel framework | C++17 and OpenMPI                          |
| Benchmark device   | CPU                                        |
| Dataset            | MOT17-derived sequences                    |
| Main process count | 12 MPI processes                           |
| Main long-run grid | 5x4 regions                                |
| Scheduler          | Static block-cyclic task assignment        |
| Placement variants | Uniform 4/4/4 and weighted 3/6/3, 4/6/2    |

## 4.4 Metrics

The report uses the following metrics from the project reports and generated CSV design:

- serial-versus-MPI correctness: mismatched frames and count error;
- detector accuracy: MAE, RMSE, percentage error, exact-match ratio;
- runtime with communication: wall-clock time including coordination;
- runtime without communication: compute-oriented time reported by the scripts;
- speedup and efficiency:

$$S_p = \frac{T_1}{T_p}, \quad E_p = \frac{S_p}{p}; \quad (4.1)$$

- idle-gap indicator for load-balance analysis.

## Chapter 5

# Results and Discussion

### 5.1 Detector Accuracy Under Tile Splitting

Table 5.1 compares detector accuracy against MOT17 ground-truth counts for the normal single-machine YOLO baseline and for different MPI tile-splitting configurations. The purpose of this experiment is to check whether spatial tiling changes the final detection accuracy, not only whether the MPI implementation is faster.

Table 5.1: YOLO accuracy comparison across baseline and tile-splitting configurations

| Configuration              | Grid | Frames | Runtime<br>(s) | MAE   | RMSE  | Percent<br>err. | Pred.<br>avg. |
|----------------------------|------|--------|----------------|-------|-------|-----------------|---------------|
| Single-machine<br>baseline | 1x1  | 300    | –              | 7.983 | 8.269 | 0.490           | 8.223         |
| MPI tiled                  | 2x2  | 300    | –              | 7.983 | 8.269 | 0.490           | 8.223         |
| MPI tiled                  | 4x3  | 300    | –              | 8.223 | 8.517 | 0.504           | 7.977         |
| MPI tiled                  | 5x4  | 300    | 54.764         | 8.383 | 8.682 | 0.514           | 7.812         |

The runtime entry is included only where the repository contains an official measurement for the same 300-frame workload. The 2x2 configuration is effectively equivalent to the single-machine baseline, so moderate spatial tiling does not change detector behavior. Finer grids introduce a small degradation: the 4x3 configuration is about 3 percent worse than baseline, and the 5x4 configuration is about 5 percent worse. This suggests that aggressive tile splitting can harm accuracy through border effects and imperfect cross-tile merging, even when it creates more parallel work.

### 5.2 Input-Size Selection

The compact MOT17 subset is useful for quick runs but does not reach the required two-to-three-minute runtime. Table 5.2 also reports the total number of frame-tile tasks and the wall-clock time per task, making it easier to compare efficiency across input sizes. The long-sequence experiment selects  $N = 600$  frames because the 12-process wall-clock runtime is 123.667 seconds.

Table 5.2: YOLO input-size selection

| Frames | $P$ | Grid | Tasks | Runtime<br>with<br>comm. (s) | Runtime<br>without<br>comm. (s) | ms/task | Selection note                  |
|--------|-----|------|-------|------------------------------|---------------------------------|---------|---------------------------------|
| 30     | 12  | 4x3  | 360   | 9.165                        | 2.999                           | 25.46   | Compact quick test              |
| 60     | 12  | 4x3  | 720   | 12.570                       | 6.801                           | 17.46   | Compact quick test              |
| 100    | 12  | 4x3  | 1200  | 17.776                       | 11.596                          | 14.81   | Compact quick test              |
| 150    | 12  | 4x3  | 1800  | 23.585                       | 17.103                          | 13.10   | Compact quick test              |
| 300    | 12  | 5x4  | 6000  | 54.764                       | 49.156                          | 9.13    | Long sequence                   |
| 600    | 12  | 5x4  | 12000 | 123.667                      | 114.352                         | 8.97    | Selected $N$                    |
| 837    | 12  | 5x4  | 16740 | 146.316                      | 139.136                         | 8.74    | Best ms/task, longer validation |
| 1200   | 12  | 5x4  | 24000 | 129.852                      | 128.122                         | 5.41    | $2N$ , weighted run             |

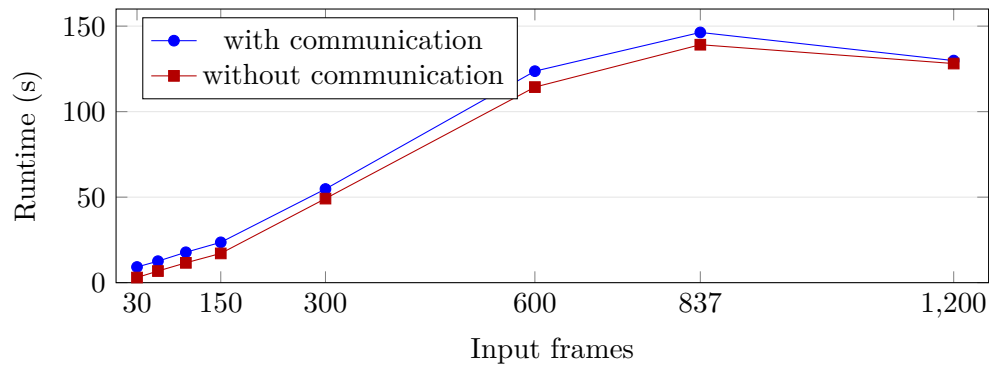


Figure 5.1: Runtime versus input size. The 600-frame workload is the first measured case inside the required 120–180 second interval.

The time-per-task column shows that very small inputs have high overhead because startup, synchronization, and aggregation are amortized over too few tasks. The 600-frame run is selected as  $N$  because it is the first long-sequence case that satisfies the required 120–180 second runtime window. The 1200-frame row is the  $2N$  speedup workload.

### 5.3 Granularity and Load Balance

Table 5.3 summarizes the official granularity experiment using weighted 6/4/2 placement. The weighted placement keeps the idle-gap indicator at 0.235, within the 25 percent course criterion. Increasing the number of regions still increases task count and communication, so granularity remains a performance trade-off even when load balance is acceptable.

Table 5.3: YOLO granularity and load balance with weighted 6/4/2 placement

| Grid | $P$ | Tasks | Max<br>comp. (s) | Avg<br>comp. (s) | Total<br>comm. (s) | Idle gap | Balance  |
|------|-----|-------|------------------|------------------|--------------------|----------|----------|
| 1x1  | 12  | 150   | 0.731            | 0.483            | 10.674             | 0.235    | Balanced |
| 2x2  | 12  | 600   | 3.155            | 2.205            | 16.306             | 0.235    | Balanced |
| 4x3  | 12  | 1800  | 9.266            | 6.375            | 34.881             | 0.235    | Balanced |
| 5x4  | 12  | 3000  | 13.733           | 9.133            | 51.790             | 0.235    | Balanced |

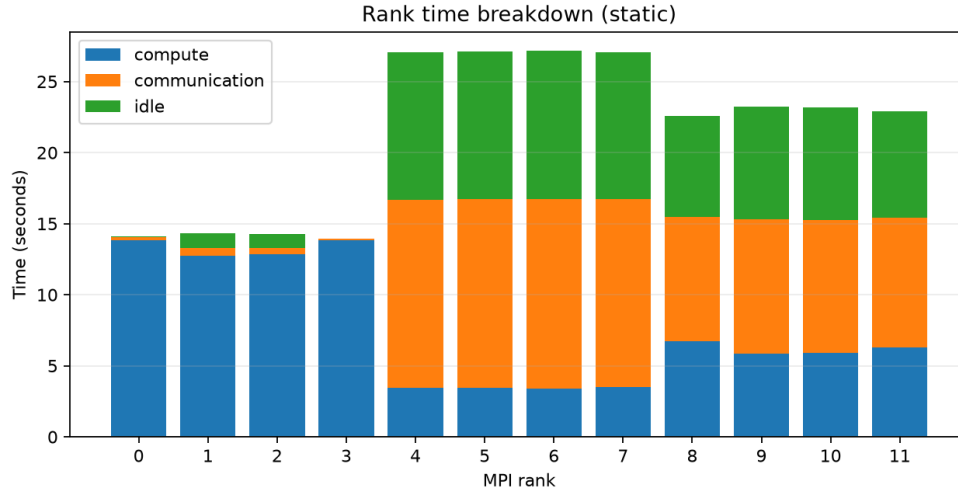


Figure 5.2: Per-process timing breakdown for the granularity experiment. Each bar is one MPI process, with compute, communication, and idle time stacked in the same column.

The result shows that weighted placement can satisfy the idle-gap criterion, but simply making tasks finer is not enough to guarantee faster execution. Finer tiles create more scheduling opportunities, but they also increase intermediate detections, communication, and master-side merging. Figure 5.2 shows why the load-balance check must inspect process-level timing rather than only total runtime.

## 5.4 Speedup and Efficiency

The speedup experiment uses the selected input size doubled to  $2N = 1200$  frames and applies weighted static placement for the multi-rank runs. This keeps the process distribution consistent with the heterogeneous cluster: stronger machines receive more ranks, while the slower machine receives fewer ranks.

Table 5.4: YOLO speedup on 1200 frames with weighted placement

| $P$ | Runtime with<br>comm. (s) | Runtime<br>without<br>comm. (s) | Wall<br>speedup | Efficiency | Note               |
|-----|---------------------------|---------------------------------|-----------------|------------|--------------------|
| 1   | 345.677                   | 342.467                         | 1.000           | 1.000      | Baseline           |
| 2   | 201.111                   | 205.100                         | 1.719           | 0.859      | Weighted placement |
| 4   | 178.238                   | 181.289                         | 1.939           | 0.485      | Weighted placement |
| 8   | 146.212                   | 146.204                         | 2.364           | 0.296      | Weighted placement |
| 12  | 129.852                   | 128.122                         | 2.662           | 0.222      | Weighted 4/6/2     |

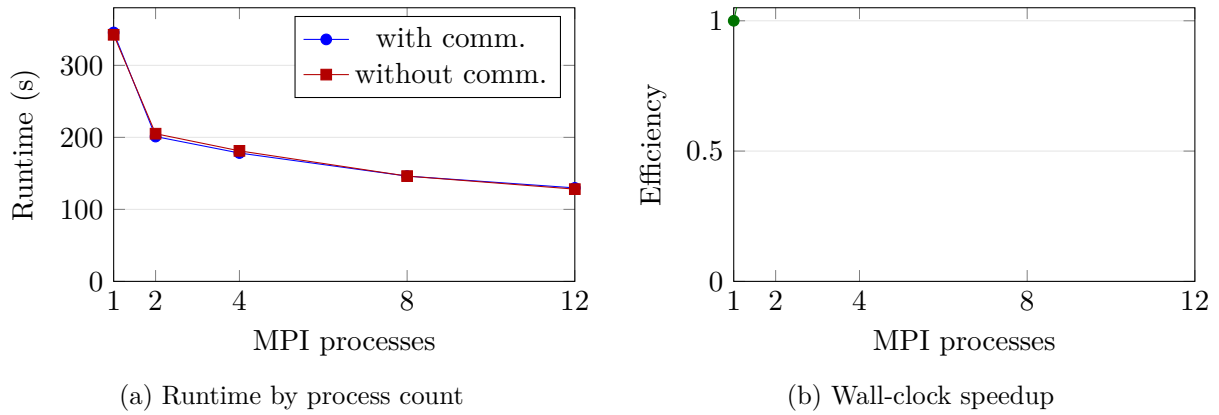


Figure 5.3: Runtime and speedup trends for the 1200-frame weighted-placement run.

The speedup is measurable but not linear. Under weighted placement, the wall-clock speedup rises from 1.719x at two processes to 2.662x at twelve processes. The 12-process computation-only speedup is approximately 2.673x. The improvement comes from reducing idle gap and assigning fewer ranks to the slower node.

## 5.5 Weighted Static Placement

The equal 4/4/4 placement does not meet the 25 percent idle-gap criterion. The best measured weighted placement assigns four ranks to the master, six to node1, and two to node2.

Table 5.5: Weighted static placement on heterogeneous cluster

| Placement | Distribution | Runtime<br>with comm.<br>(s) | Runtime<br>without<br>comm. (s) | Idle gap | Balance      |
|-----------|--------------|------------------------------|---------------------------------|----------|--------------|
| Uniform   | 4/4/4        | 40.619                       | 36.522                          | 0.466    | Not balanced |
| Weighted  | 3/6/3        | 34.712                       | 30.339                          | 0.371    | Not balanced |
| Weighted  | 4/6/2        | 29.581                       | 27.484                          | 0.235    | Balanced     |

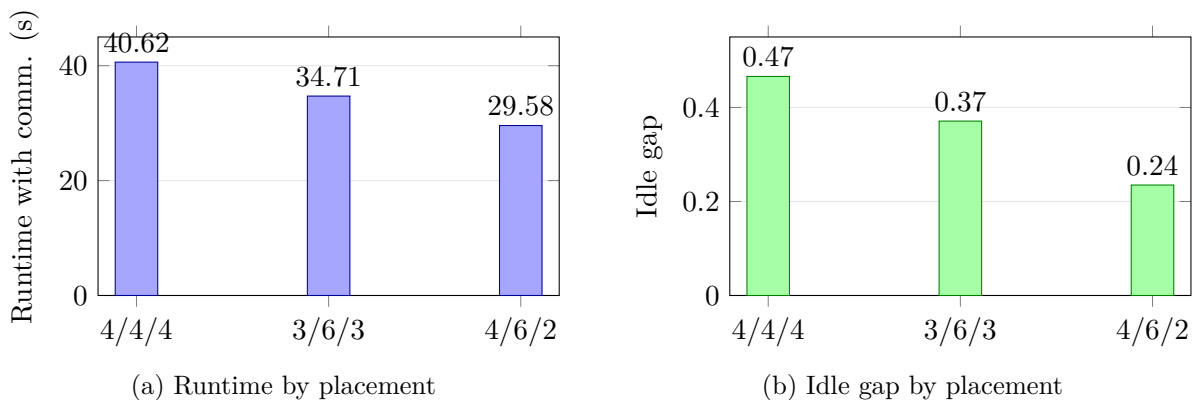


Figure 5.4: Weighted placement reduces both runtime and idle-gap imbalance.



This result supports the claim that processor assignment matters on a heterogeneous physical cluster. Equal rank placement gives every host the same number of processes, but the machines do not have equal throughput. Weighted placement reduces runtime and satisfies the load-balance criterion.

## 5.6 YOLO Parallel Correctness

Table 5.6 reports the serial-versus-MPI correctness result recorded in the repository report. The MPI run produced the same frame counts as the serial baseline on the tested sequence.

Table 5.6: YOLO serial-versus-MPI correctness

| Result | Frames | Mismatched<br>frames | Max count<br>error | Mean error | Evidence          |
|--------|--------|----------------------|--------------------|------------|-------------------|
| Passed | 30     | 0                    | 0                  | 0.000      | repository report |

This validates the parallel pipeline for the tested configuration. It indicates that task splitting, MPI result collection, coordinate remapping, duplicate removal, and final counting preserve the serial pipeline output.

## 5.7 Summary of Findings

The measured YOLO results support the main task-parallel narrative. The implementation preserves serial output, reaches the required input-size runtime, shows measurable speedup, and demonstrates that task granularity and rank placement strongly affect performance. The strongest measured improvement in load balance comes from weighted static placement, which reduces the idle-gap indicator from 0.466 to 0.235.

## Chapter 6

# Conclusion

This report presents the final content for *Parallelization Methods for CNN-Based Computer Vision Inference* using evidence from the `yolo-mpi-people-count` repository. The completed YOLO method demonstrates task-level parallelism with hybrid temporal-spatial decomposition, 1D task mapping, static block-cyclic scheduling, rank-local detector execution, and centralized detection merging.

The measured YOLO results are strong enough for the main experimental narrative. The serial-versus-MPI correctness test passes with zero mismatched frames. The selected 600-frame workload runs in 123.667 seconds with 12 processes, satisfying the required two-to-three-minute range. The official 1200-frame weighted 4/6/2 run reaches 129.852 seconds and 2.662x wall-clock speedup. Weighted static placement improves the idle-gap indicator to 0.235 and satisfies the 25 percent load-balance criterion.

The detector accuracy result should be interpreted separately from parallel correctness. YOLO undercounts crowded MOT17 frames, with MAE 7.983 on the recorded 300-frame accuracy experiment. This is a model limitation, not evidence that MPI changed the pipeline output.

Overall, the project covers the core course concepts through one coherent method: decomposition, mapping, communication topology, scheduling, load balancing, correctness verification, input-size selection, and speedup measurement.

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